

Business Intelligence Mini-Project

Customer Shopping Behavior Analysis

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1. Introduction

A clothing retailer collects thousands of small decisions every day: a customer scrolls past a jacket, hesitates over a discount code, picks a size, checks out. On their own, these moments say very little. Stacked next to each other by the thousand, though, they start to reveal something closer to a pattern: who buys, when they buy, what pulls them back, and what a subscription or a promo code is actually worth to the business behind the scenes.

This project works through one such stack: a transactional dataset covering 3,900 purchases across a clothing company's product catalogue. The aim was not simply to describe the data, but to move it through a full analytics pipeline, from raw CSV to a decision-ready dashboard, and to come out the other side with a handful of concrete, defensible recommendations about subscriptions, discounting, and where the company's marketing budget is best spent.

The work was carried out in three stages. First, the dataset was cleaned and reshaped in Python. Second, it was loaded into PostgreSQL, where a set of business questions were answered directly in SQL. Third, the cleaned data was brought into Power BI to build an interactive dashboard that a non-technical stakeholder could use to explore the same findings without touching a single line of code.

2. Implementation

2.1. Data Gathering

The dataset was sourced as a single CSV file, containing 3,900 rows and 18 columns. It combines three broad families of information: customer demographics (age, gender, location, subscription status), purchase details (item purchased, category, purchase amount, season, size, colour), and shopping behaviour (discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type). The only meaningful gap in the raw data was in the Review Rating column, which was missing 37 values out of 3,900, a small enough share to repair rather than discard.

2.2. Data Preparation

Cleaning and reshaping the dataset was done in Python, using pandas for the heavy lifting. The process followed a fairly standard path, but each step mattered for what came later:

- Loading and inspection: the CSV was read into a DataFrame, then examined with `.info()` and `.describe()` to confirm structure, types, and the shape of the missing data.
- Missing data handling: the 37 missing review ratings were imputed using the median rating within each product category, rather than a single global median, so that a gap in the Boots category was filled with a Boots-appropriate value instead of an average across the whole catalogue.
- Column standardisation: every column name was converted to snake_case, mostly to keep the later SQL queries readable and consistent.
- Feature engineering: an age_group column was created by binning raw ages into cohorts (Young Adult, Adult, Middle-aged, Senior), and a purchase_frequency_days column was derived from the existing

purchase-frequency field so that recurrence could be measured in a consistent unit.

- Consistency check: discount_applied and promo_code_used turned out to be redundant, tracking essentially the same behaviour, so promo_code_used was dropped to avoid double counting the same signal in later analysis.
- Database integration: once cleaned, the DataFrame was loaded into a PostgreSQL database, which became the working surface for the SQL analysis in the next section.

2.3. Data Analysis Using SQL

With the cleaned data sitting in PostgreSQL, ten business questions were answered directly in SQL. The queries themselves ranged from simple aggregations to segment-style classifications, and the results below are what actually shaped the recommendations in Section 4.

Revenue by Gender

Comparing total revenue by gender surfaced a lopsided split: male customers generated roughly twice the revenue of female customers over the period covered by

	gender text	revenue numeric
1	Female	75191
2	Male	157890

the dataset.

High-Spending Discount Users

This query isolated customers who used a discount but still spent above the dataset-wide average purchase amount, a group worth watching because they are not simply chasing the lowest price. 839 customers met that criterion; a sample of the results

customer_id bigint	purchase_amount bigint
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

is shown below.

Top 5 Products by Average Rating Shipping Type Comparison

item_purchased text	Average Product Rating numeric
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Standard and Express shipping were compared on average purchase amount. The gap is narrow, roughly two dollars, which suggests shipping speed is not, by itself, a strong lever on basket size.

shipping_type text	round numeric
Standard	58.46
Express	60.48

Subscribers vs. Non-Subscribers

Subscribed customers spend at almost the same average rate as non-subscribers, but there are far fewer of them, which caps their contribution to total revenue even though the loyalty logic behind a subscription programme suggests they should be the more valuable segment over time.

subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

Discount-Dependent Products

Some products lean on discounting far more than others. The five items below had close to half of their purchases made under a discount, which flags them as candidates for margin review.

item_purchased text	discount_rate numeric
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Customer Segmentation

Customers were classified into New, Returning, and Loyal segments based on purchase history. The overwhelming majority already sit in the Loyal segment, which is encouraging, but also means the New and Returning segments (only 784 customers combined) are the ones with the most room to grow.

customer_segment text	Number of Customers bigint
Loyal	3116
New	83
Returning	701

Top 3 Products per Category

category text	item_purchased text	total_purchases bigint	rank bigint
Accessori...	Jewelry	171	1
Accessori...	Belt	161	2
Accessori...	Sunglasses	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

Repeat Buyers and Subscriptions

Checking whether customers with more than five purchases are more likely to subscribe gave a clear answer: no. 2,518 repeat buyers are not subscribed, against only 958 who are, meaning loyalty in behaviour is not currently converting into loyalty in enrolment.

subscription_status 	repeat_buyers 
text	bigint
No	2518
Yes	958

Revenue by Age Group

age_group 	total_revenue 
text	numeric
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

3. Data Visualization

The cleaned dataset and the SQL results were brought together into a single Power BI dashboard, built so that someone outside the analysis, a category manager or a marketing lead, could filter by subscription status, gender, category, or shipping type and see the numbers move in front of them rather than read them off a table.

At a glance, the dashboard anchors on three headline figures: 3,900 customers, an average purchase amount of \$59.76, and an average review rating of 3.75 out of 5. Beneath that, a donut chart breaks the base down by subscription status (27% subscribed, 73% not), while paired bar charts set revenue against sales volume, first by product category, then by age group, so that a category or a generation that sells a lot but earns comparatively little becomes visible at a glance rather than buried in a spreadsheet.



4. Business Recommendations

Taken together, the SQL analysis and the dashboard point toward five concrete moves:

- Boost subscriptions. With subscribers making up only 27% of the customer base despite spending at nearly the same rate as everyone else, there is a clear, largely untapped source of revenue in simply growing that number, through more visible benefits or a more deliberate sign-up push at checkout.
- Build a loyalty programme aimed at repeat buyers. 2,518 customers already purchase often enough to qualify as repeat buyers without being subscribed. Converting even a fraction of that group would do more for revenue than acquiring new customers from scratch.
- Revisit the discount policy on a short list of products. Hats, sneakers, coats, sweaters, and pants are all sold at a discount close to half the time, which is worth checking against their margins before assuming the discount is paying for itself.
- Lean into what already rates well. Gloves, sandals, boots, hats, and skirts top the ratings table and deserve more visible placement in campaigns and on the storefront rather than competing for attention with lower-rated items.
- Focus marketing spend where the returns are: on Young Adult and Middle-aged customers, who currently generate the most revenue, and on Express shipping users, who tend to purchase slightly more per order than Standard shipping customers.

5. Conclusion

What this project ultimately shows is less a single insight than a set of small, connected ones: a subscription base that is smaller than it should be given how well subscribers actually spend, a loyal customer segment that already exists but has not been rewarded for its loyalty, and a handful of products whose sales numbers may be flattered by discounting more than the raw figures let on.

None of these conclusions required particularly exotic tooling: a pandas cleaning pass, a set of SQL queries answering plain business questions, and a Power BI dashboard to make the results legible to someone who was not in the room for the analysis. That, more than any single chart, is probably the real takeaway of this project: a fairly ordinary pipeline, applied carefully, is usually enough to turn a spreadsheet of transactions into something an actual business decision can rest on.